Kishinchand Chellaram College, Mumbai – 20.

FY/SY /TY B. Sc. (I. T) Semester_____

Practical 1

Aim: Implementing virtual environment for making an object jump.

Step No.	Step
1.	Open Unity3D
2.	Click on New Project --> Select 'All Templates' --> Select 3D Core --> Give a Project Name --> Click on Create Project
3.	Double Click on the Project Created
4.	Select Game Object --> 3D Object --> Plane
5.	Select Game Object --> 3D Object --> Sphere
6.	Now Click on 'Add Component' --> Select 'Rigid Body'
7.	Now Click on 'Add Component' --> Type a **new name** --> Select Script --> Select Create and Add
8.	Double Click on the Script File Created
9.	Type the code in the file that opens:

```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;

public class Sample : MonoBehaviour
{

    //Rigidbody rb;
    Rigidbody rb2;
    // Start is called before the first frame update
    void Start()
    {
      //  rb = gameObject.GetComponent<Rigidbody>();
        rb2 = gameObject.GetComponent<Rigidbody>();
    }

    // Update is called once per frame
    void Update()
    {
        if(Input.GetKeyDown(KeyCode.Space))
        {
            rb2.AddForce(Vector3.up * 10,ForceMode.Impulse);
        }
    }
}
```

10.	Save and Go Back TO Unity3D
11.	Click on the 'Green Play Button' or 'Run'

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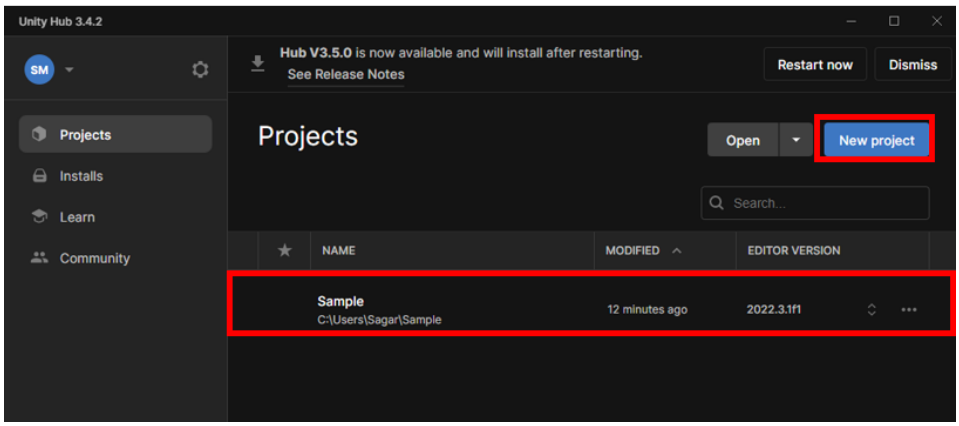
Screenshots:

Step 1 to 3



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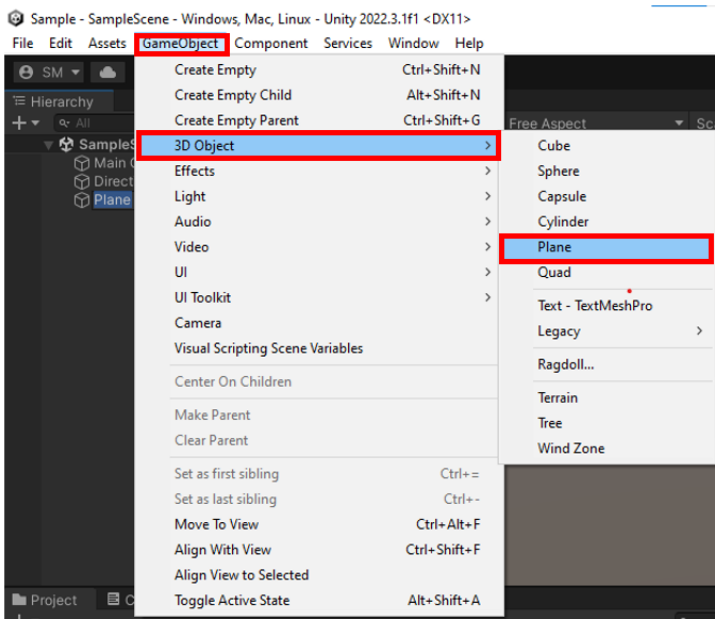

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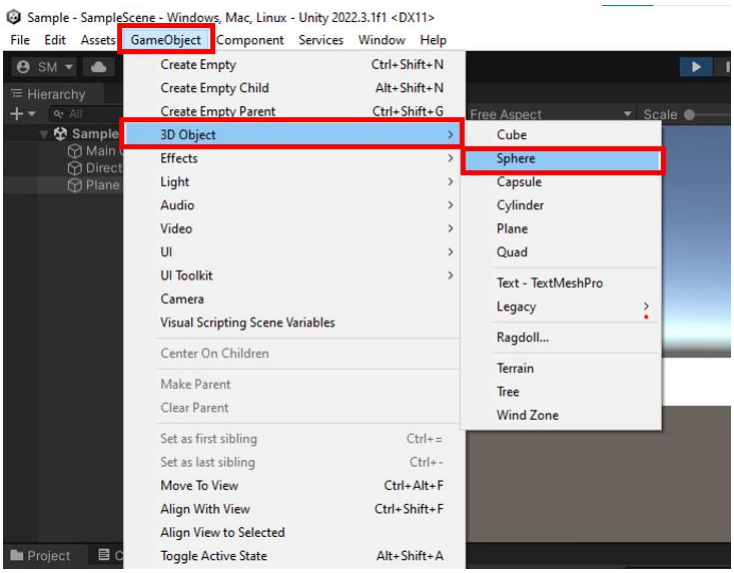

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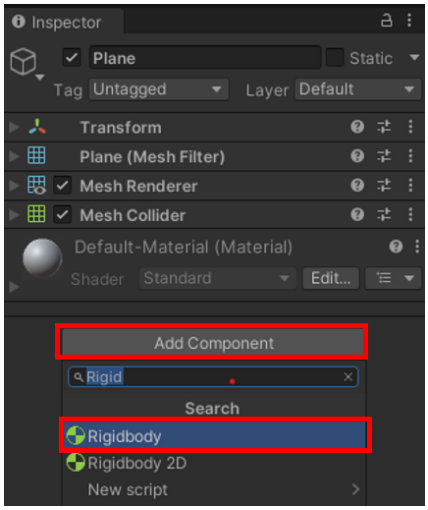

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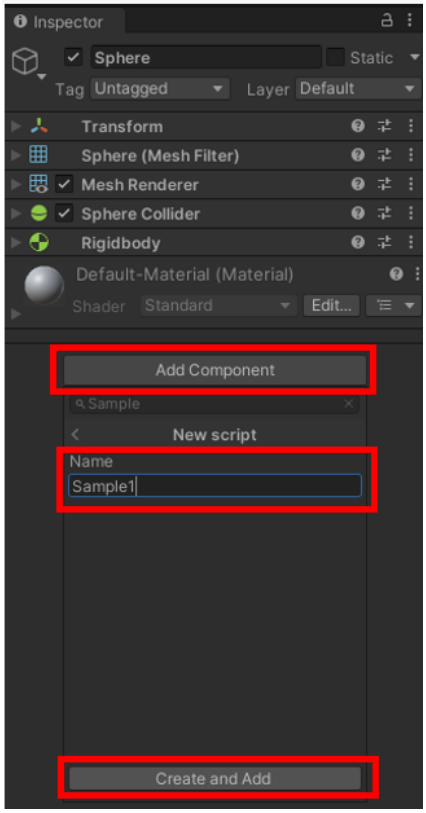

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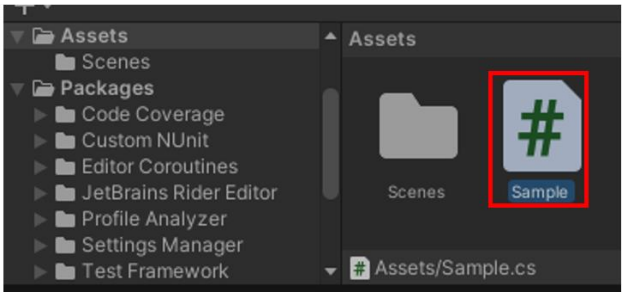

Step 9:

C# Code

```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;

public class Sample : MonoBehaviour
{

   //Rigidbody rb;
   Rigidbody rb2;
   // Start is called before the first frame update
   void Start()
   {
    //  rb = gameObject.GetComponent<Rigidbody>();
      rb2 = gameObject.GetComponent<Rigidbody>();
   }

   // Update is called once per frame
   void Update()
   {
      if(Input.GetKeyDown(KeyCode.Space))
      {
         rb2.AddForce(Vector3.up * 10,ForceMode.Impulse);
      }
   }
}
```

Step 10:

Run the file in Unity3D and see the output.

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Practical No. 2

Aim: Implementing virtual environment for moving and making the object jump

Step No.	Step
1.	Open Unity3D
2.	Click on New Project --> Select 'All Templates' --> Select 3D Core --> Give a Project Name --> Click on Create Project
3.	Double Click on the Project Created
4.	Select Game Object --> 3D Object --> Plane
5.	Select Game Object --> 3D Object --> Sphere
6.	Now Click on 'Add Component' --> Select 'Rigid Body'
7.	Now Click on 'Add Component' --> Type a **new name** --> Select Script --> Select Create and Add
8.	Double Click on the Script File Created
9.	Type the code in the file that opens:

```
using System.Collections;
using System.Collections.Generic;
using Unity.VisualScripting;
using UnityEngine;

public class Movement : MonoBehaviour
{
    Rigidbody rb2;
    public float speed = 0.1f;
    // Start is called before the first frame update
    void Start()
    {
        rb2 = gameObject.GetComponent<Rigidbody>();
    }

    // Update is called once per frame
    void Update()
    {
        if (Input.GetKey(KeyCode.RightArrow))
        {
            transform.Translate(50f * speed * Time.deltaTime, 0, 0);
        }
        else if (Input.GetKey(KeyCode.LeftArrow))
        {
            transform.Translate(-50f * speed * Time.deltaTime, 0,
0);
        }
        else if (Input.GetKeyDown(KeyCode.Space))
        {
            rb2.AddForce(Vector3.up * 5, ForceMode.Impulse);
        }
    }
}
```

10.	Save and Go Back TO Unity3D
11.	Click on the 'Green Play Button' or 'Run'

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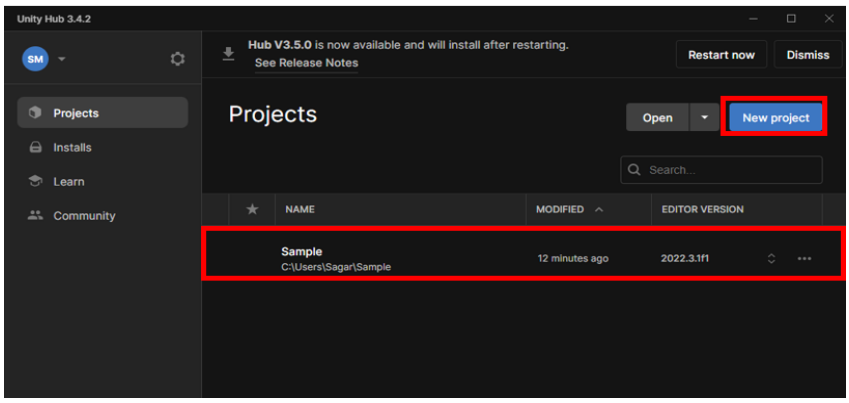
Screenshots:

Step 1 to 3



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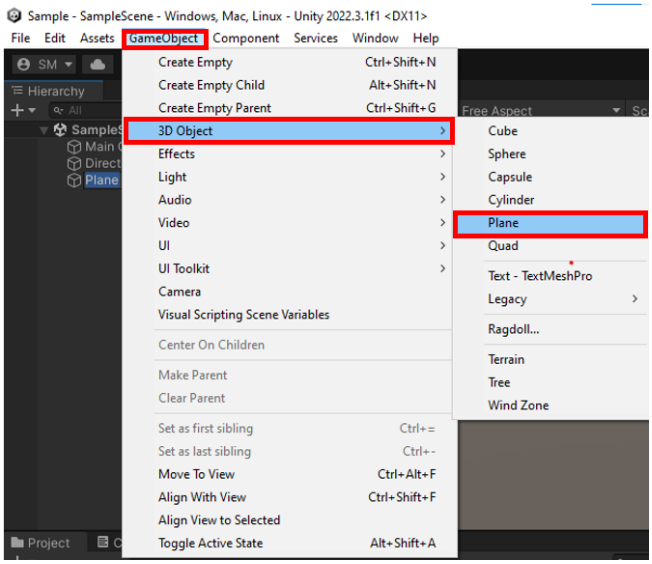

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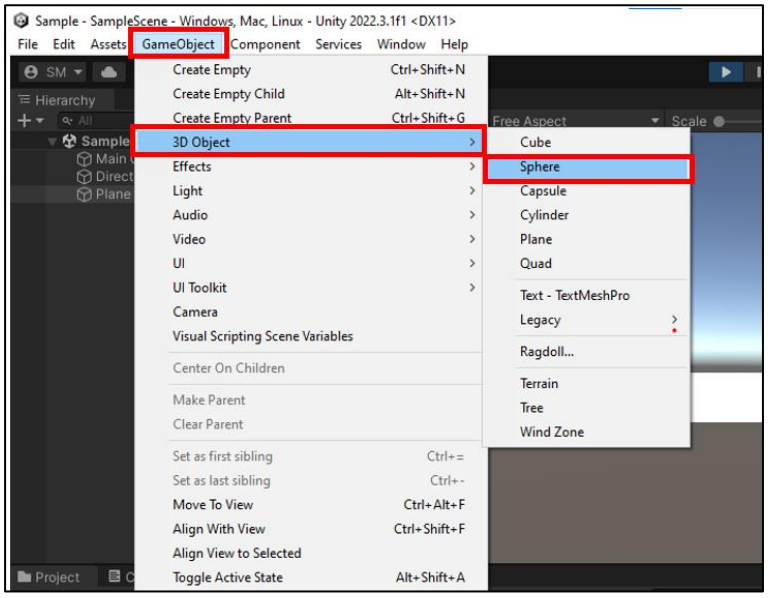

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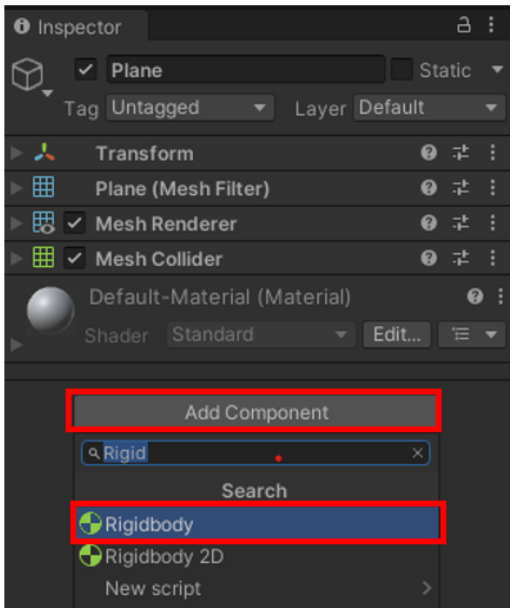

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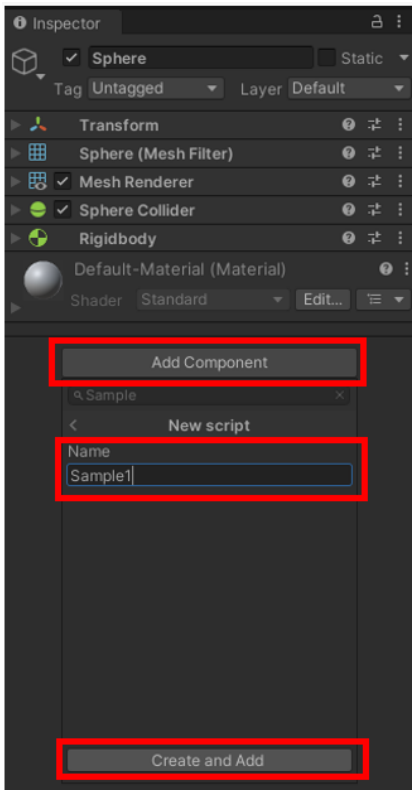

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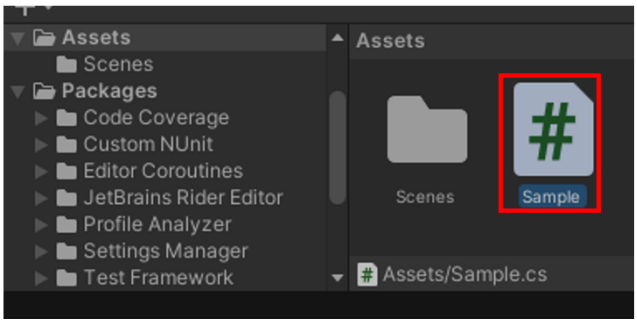

Step 9:

C# Code

```
using System.Collections;
using System.Collections.Generic;
using Unity.VisualScripting;
using UnityEngine;

public class Movement : MonoBehaviour
{
    Rigidbody rb2;
    public float speed = 0.1f;
    // Start is called before the first frame update
    void Start()
    {
        rb2 = gameObject.GetComponent<Rigidbody>();
    }

    // Update is called once per frame
    void Update()
    {
        if (Input.GetKey(KeyCode.RightArrow))
        {
            transform.Translate(50f * speed *
Time.deltaTime, 0, 0);
        }
        else if (Input.GetKey(KeyCode.LeftArrow))
        {
            transform.Translate(-50f * speed *
Time.deltaTime, 0, 0);
        }
        else if (Input.GetKeyDown(KeyCode.Space))
        {
            rb2.AddForce(Vector3.up * 5,
ForceMode.Impulse);
        }
    }
}
```

Step 10:

Run the file in Unity3D and see the output.

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Practical 3

Aim: Implementing virtual environment for moving the object in all directions

Step No.	Step
1.	Open Unity3D
2.	Click on New Project --> Select 'All Templates' --> Select 3D Core --> Give a Project Name --> Click on Create Project
3.	Double Click on the Project Created
4.	Select Game Object --> 3D Object --> Plane
5.	Select Game Object --> 3D Object --> Sphere
6.	Now Click on 'Add Component' --> Select 'Rigid Body'
7.	Now Click on 'Add Component' --> Type a **new name** --> Select Script --> Select Create and Add
8.	Double Click on the Script File Created
9.	Type the code in the file that opens:

```
using System.Collections;
using System.Collections.Generic;
using Unity.VisualScripting;
using UnityEngine;

public class Movement : MonoBehaviour
{
    Rigidbody rb2;
    public float speed = 2f;
    // Start is called before the first frame update
    void Start()
    {
       rb2 = gameObject.GetComponent<Rigidbody>();
    }

    // Update is called once per frame
    void Update()
    {
        if (Input.GetKey(KeyCode.RightArrow))
        {
            transform.Translate(50f * speed * Time.deltaTime, 0, 0);
        }
        else if (Input.GetKey(KeyCode.LeftArrow))
        {
            transform.Translate(-50f * speed * Time.deltaTime, 0, 0);
        }
        else if (Input.GetKeyDown(KeyCode.Space))
        {
          rb2.AddForce(Vector3.up * 5, ForceMode.Impulse);
        }
        else if (Input.GetKey(KeyCode.DownArrow))
        {
            transform.Translate(Vector3.forward * Time.deltaTime);
        }
        else if (Input.GetKey(KeyCode.UpArrow))
        {
            this.transform.Translate(Vector3.back * Time.deltaTime);
        }
    }
}
```

10.	Save and Go Back TO Unity3D
11.	Click on the 'Green Play Button' or 'Run'

Kishinchand Chellaram College, Mumbai – 20.

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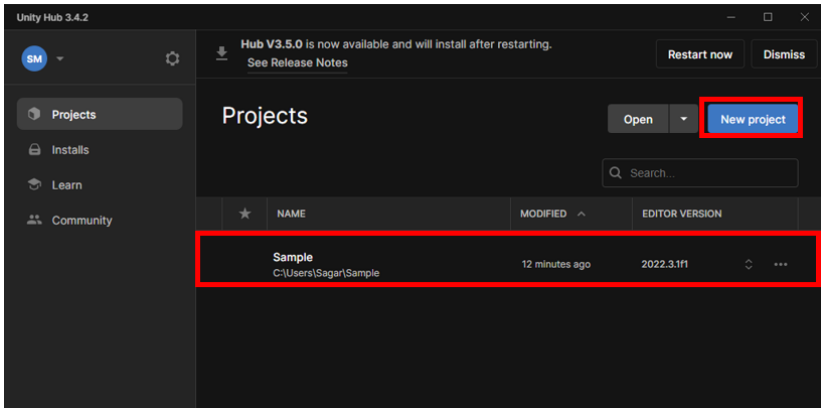
Screenshots:

Step 1 to 3



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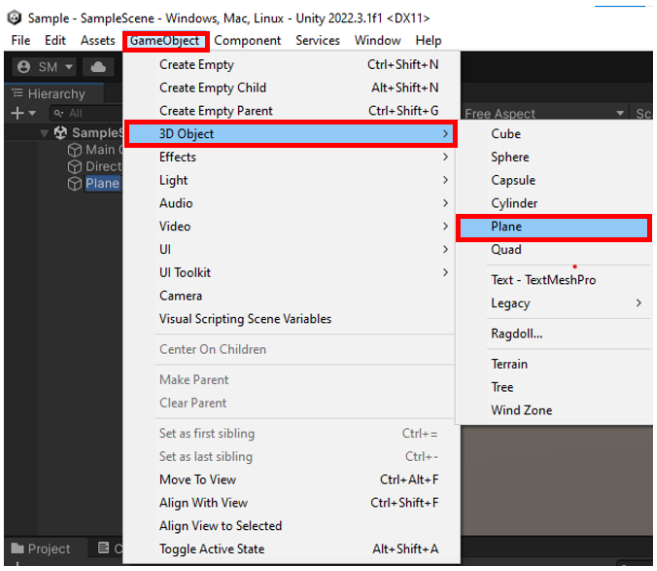

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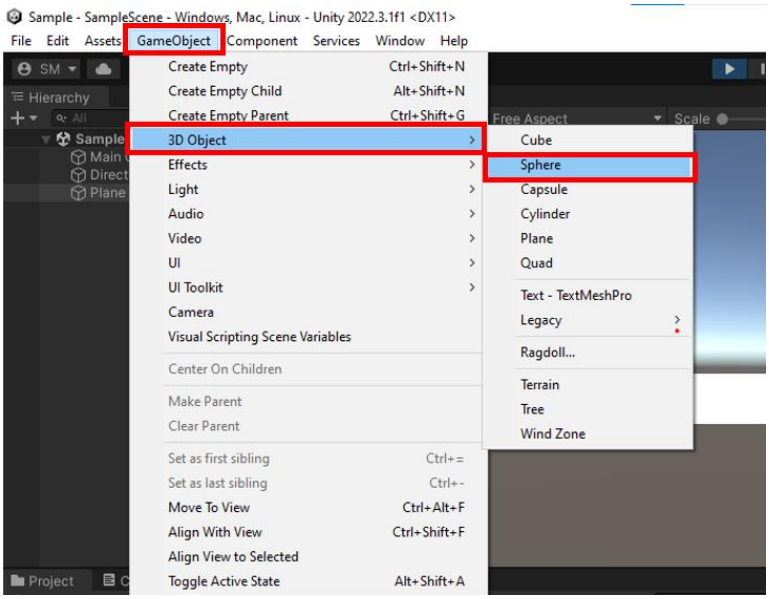

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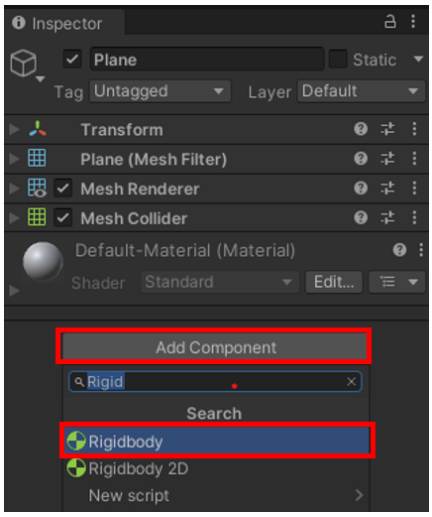
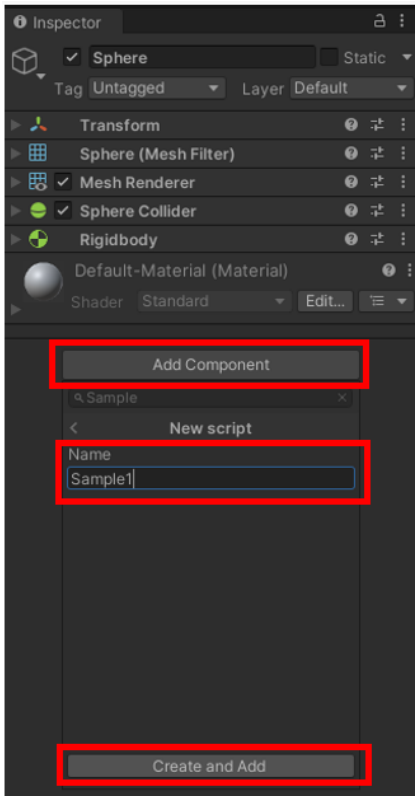
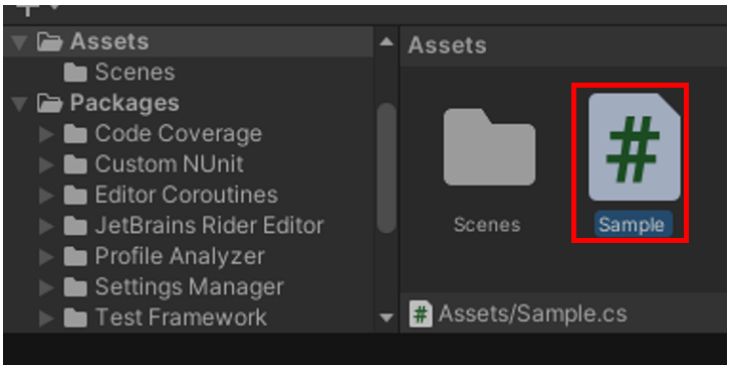


Step 7:



Step 8:



Step 9:

C# Code

```
using System.Collections;
using System.Collections.Generic;
using Unity.VisualScripting;
using UnityEngine;

public class Movement : MonoBehaviour
{
    Rigidbody rb2;
    public float speed = 2f;
    // Start is called before the first frame update
    void Start()
    {
       rb2 = gameObject.GetComponent<Rigidbody>();
    }

    // Update is called once per frame
    void Update()
    {
        if (Input.GetKey(KeyCode.RightArrow))
        {
            transform.Translate(50f * speed *
Time.deltaTime, 0, 0);
        }
        else if (Input.GetKey(KeyCode.LeftArrow))
        {
            transform.Translate(-50f * speed *
Time.deltaTime, 0, 0);
        }
        else if (Input.GetKeyDown(KeyCode.Space))
        {
          rb2.AddForce(Vector3.up * 5, ForceMode.Impulse);
        }
        else if (Input.GetKey(KeyCode.DownArrow))
        {
            transform.Translate(Vector3.forward *
Time.deltaTime);
        }
        else if (Input.GetKey(KeyCode.UpArrow))
        {
            this.transform.Translate(Vector3.back *
Time.deltaTime);
        }
    }
}
```

Step 10:

Run the file in Unity3D and see the output.

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Practical 4

Aim: Implementing virtual environment for making the object Change Colour

Step No.	Step
1.	Open Unity3D
2.	Click on New Project --> Select 'All Templates' --> Select 3D Core --> Give a Project Name --> Click on Create Project
3.	Double Click on the Project Created
4.	Select Game Object --> 3D Object --> Plane
5.	Select Game Object --> 3D Object --> Sphere
6.	Now Click on 'Add Component' --> Select 'Rigid Body'
7.	Now Click on 'Add Component' --> Type a **new name** --> Select Script --> Select Create and Add
8.	Double Click on the Script File Created
9.	Type the code in the file that opens:

```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;

public class colourchanger : MonoBehaviour
{
    // Start is called before the first frame update
    void Start()
    {

    }

    // Update is called once per frame
    void Update()
    {
        if (Input.GetKey(KeyCode.R))
        {
            GetComponent<Renderer>().material.color = Color.red;
        }
        if (Input.GetKey(KeyCode.B))
        {
            GetComponent<Renderer>().material.color = Color.blue;
        }
        if (Input.GetKey(KeyCode.Y))
        {
            GetComponent<Renderer>().material.color = Color.yellow;
        }
        if (Input.GetKey(KeyCode.G))
        {
            GetComponent<Renderer>().material.color = Color.green;
        }

    }
}
```

10.	Save and Go Back TO Unity3D
11.	Click on the 'Green Play Button' or 'Run'

Screenshots:

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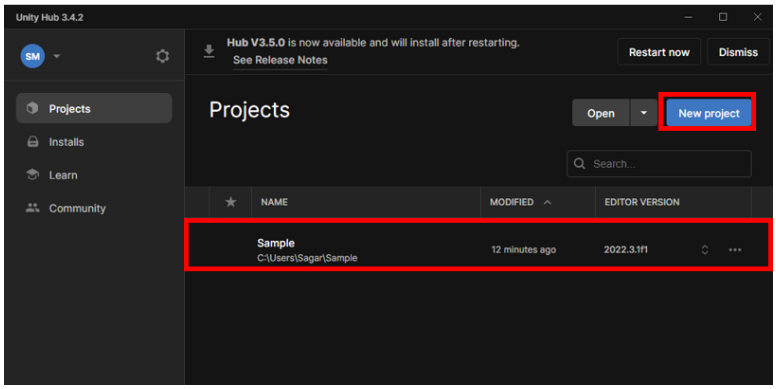
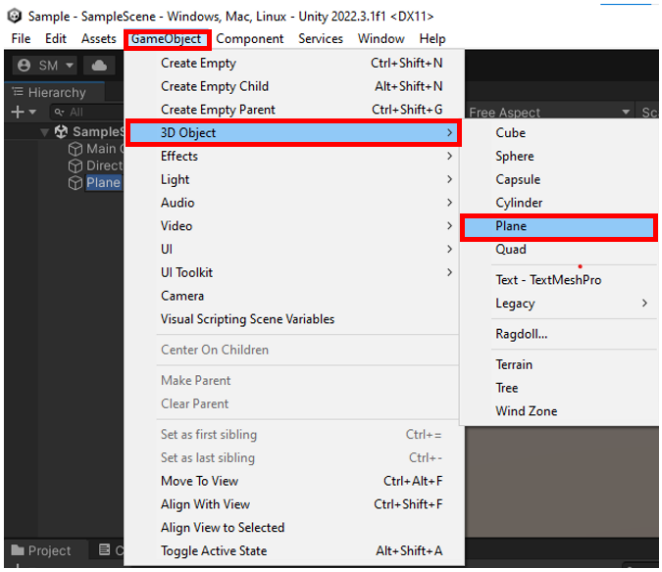
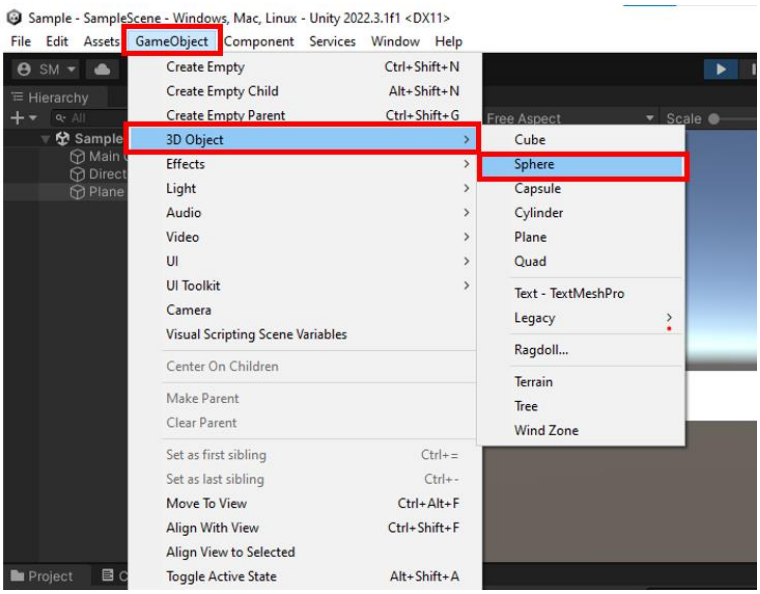
Step 1 to 3



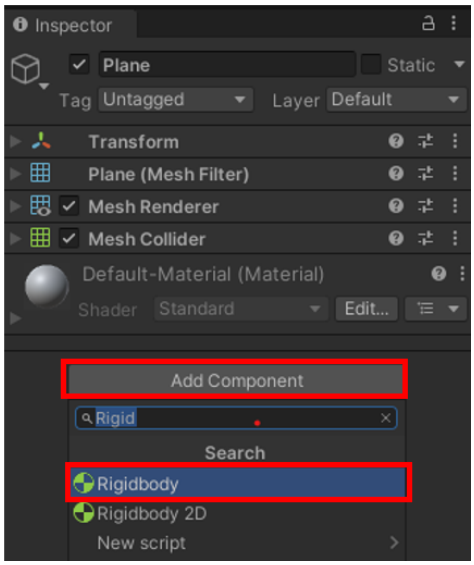
Step 4:



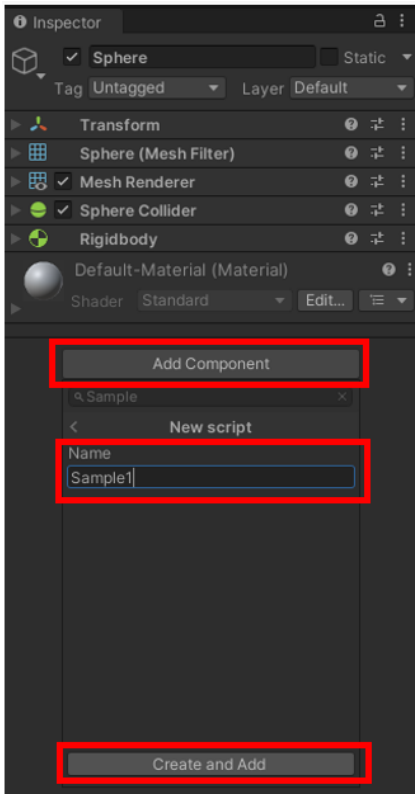
Step 5:



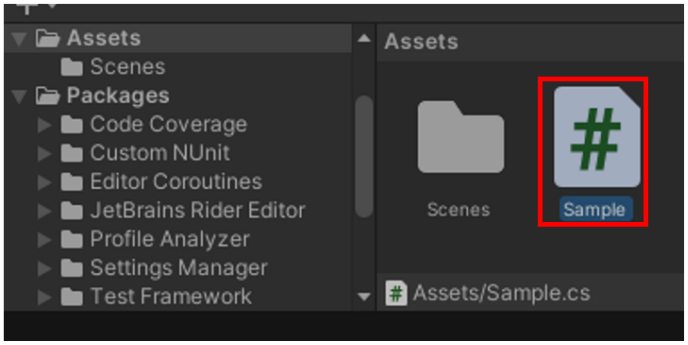
Step 6:



Step 7:



Step 8:



Step 9:

C# Code

```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;
public class colourchanger : MonoBehaviour
{
    // Start is called before the first frame update
    void Start()
    {
    }
    // Update is called once per frame
    void Update()
    {
        if (Input.GetKey(KeyCode.R))
        {
            GetComponent<Renderer>().material.color = Color.red;
        }
        if (Input.GetKey(KeyCode.B))
        {
            GetComponent<Renderer>().material.color = Color.blue;
        }
        if (Input.GetKey(KeyCode.Y))
        {
            GetComponent<Renderer>().material.color = Color.yellow;
        }
        if (Input.GetKey(KeyCode.G))
        {
            GetComponent<Renderer>().material.color = Color.green;
        }
    }
}
```

Step 10:

Run the file in Unity3D and see the output.

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Practical 5

Aim: Colour Randomizer

Step No.	Step
1.	Open Unity3D
2.	Click on New Project --> Select 'All Templates' --> Select 3D Core --> Give a Project Name --> Click on Create Project
3.	Double Click on the Project Created
4.	Select Game Object --> 3D Object --> Plane
5.	Select Game Object --> 3D Object --> Sphere
6.	Now Click on 'Add Component' --> Select 'Rigid Body'
7.	Now Click on 'Add Component' --> Type a **new name** --> Select Script --> Select Create and Add
8.	Double Click on the Script File Created
9.	Type the code in the file that opens:

```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;

public class ColorRandomizer : MonoBehaviour
{
    public float speed = 5f;
    // Start is called before the first frame update
    void Start()
    {
    }

    // Update is called once per frame
    void Update()
    {
        float h = Input.GetAxis("Horizontal");
        float v = Input.GetAxis("Vertical");
        transform.Translate(h * speed * Time.deltaTime, v * speed *
Time.deltaTime, 0);
        if (Input.GetKey(KeyCode.Space))
        {
            GetComponent<Renderer>().material.color = Random.ColorHSV(0f,
1f, 1f, 1f, 0.5f, 1f);
        }

    }
}
```

10.	Save and Go Back TO Unity3D
11.	Click on the 'Green Play Button' or 'Run'

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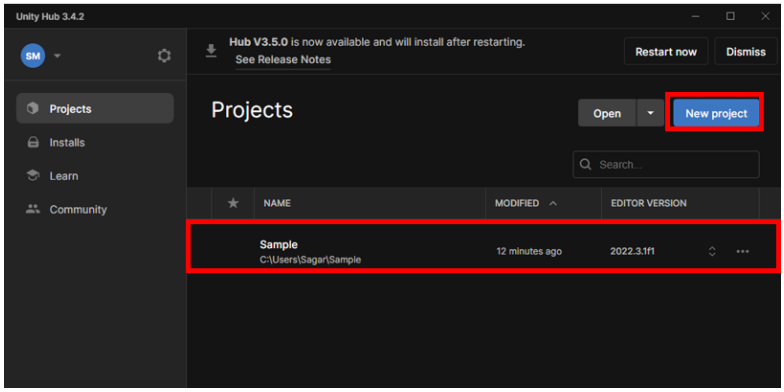
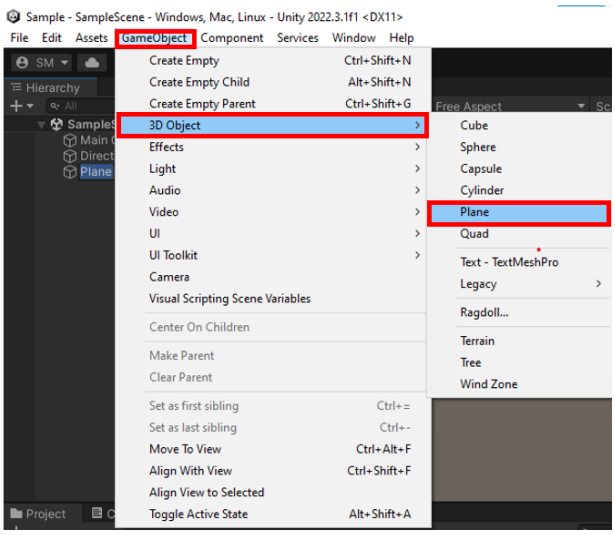
Screenshots:

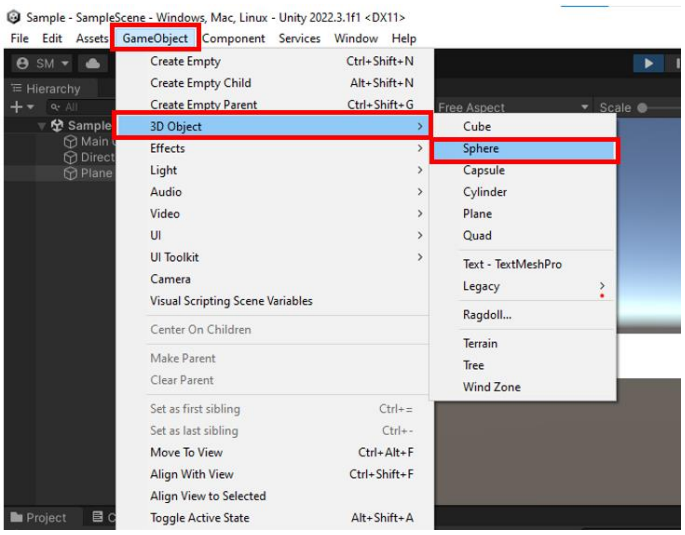
Step 1 to 3



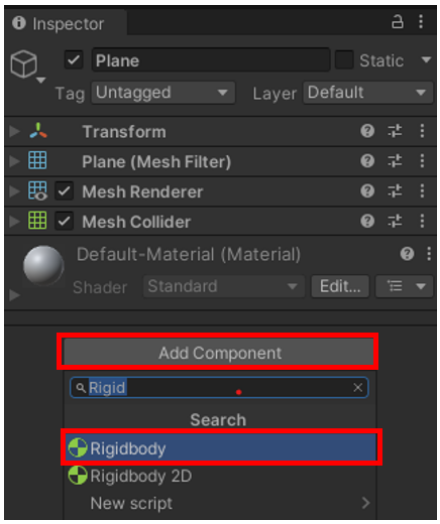
Step 4:



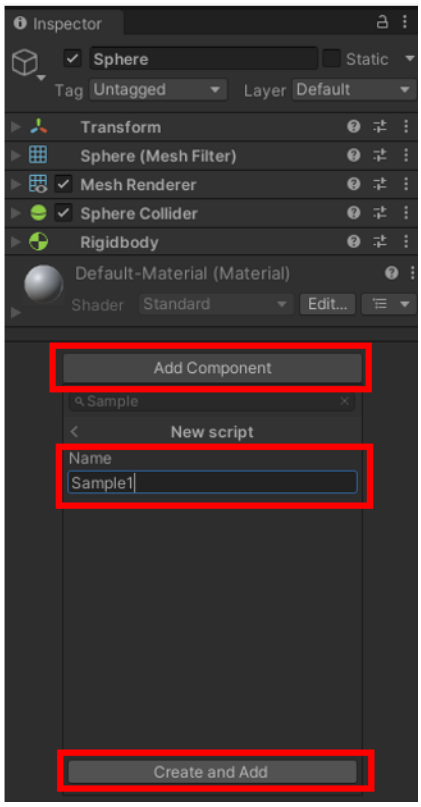
Step 5:



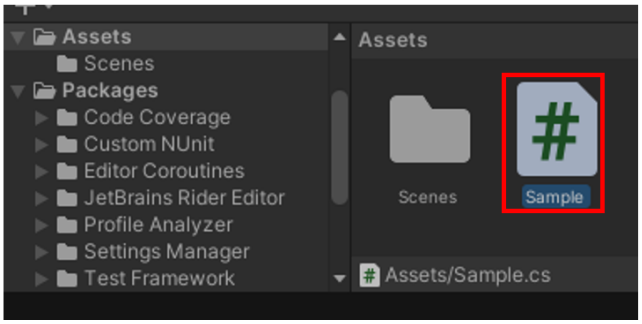
Step 6:



Step 7:



Step 8:



Step 9:

C# Code

```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;

public class ColorRandomizer : MonoBehaviour
{
    public float speed = 5f;
    // Start is called before the first frame update
    void Start()
    {
    }
    // Update is called once per frame
    void Update()
    {
        float h = Input.GetAxis("Horizontal");
        float v = Input.GetAxis("Vertical");
        transform.Translate(h * speed *
Time.deltaTime, v * speed * Time.deltaTime, 0);
        if (Input.GetKey(KeyCode.Space))
        {
            GetComponent<Renderer>().material.color =
Random.ColorHSV(0f, 1f, 1f, 1f, 0.5f, 1f);
        }
    }
}
```

Step 10:

Run the file in Unity3D and see the output

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Practical 6

Aim: Implementing virtual environment to Enabling Lights

Step No.	Step
1.	Open Unity3D
2.	Click on New Project --> Select 'All Templates' --> Select 3D Core --> Give a Project Name --> Click on Create Project
3.	Double Click on the Project Created
4.	Select Game Object --> 3D Object --> Plane
5.	Select Game Object --> 3D Object --> Sphere
6.	Now Click on 'Add Component' --> Select 'Rigid Body'
7.	Now Click on 'Add Component' --> Type a **new name** --> Select Script --> Select Create and Add
8.	Double Click on the Script File Created

9.	Type the code in the file that opens:
	
	`using System.Collections;`
	`using System.Collections.Generic;`
	`using UnityEngine;`
	
	`public class light : MonoBehaviour`
	`{`
	`  public Light myLight;`
	
	`  void Start()`
	`  {`
	`  myLight.GetComponent<Light>();`
	`  }`
	
	`  // Update is called once per frame`
	`  void Update()`
	`  {`
	`  if (Input.GetKey(KeyCode.L))`
	`  {`
	`  myLight.enabled = !myLight.enabled;`
	`  }`
	`  }`
	
	`}`

10.	Save and Go Back TO Unity3D
11.	Click on the 'Green Play Button' or 'Run'

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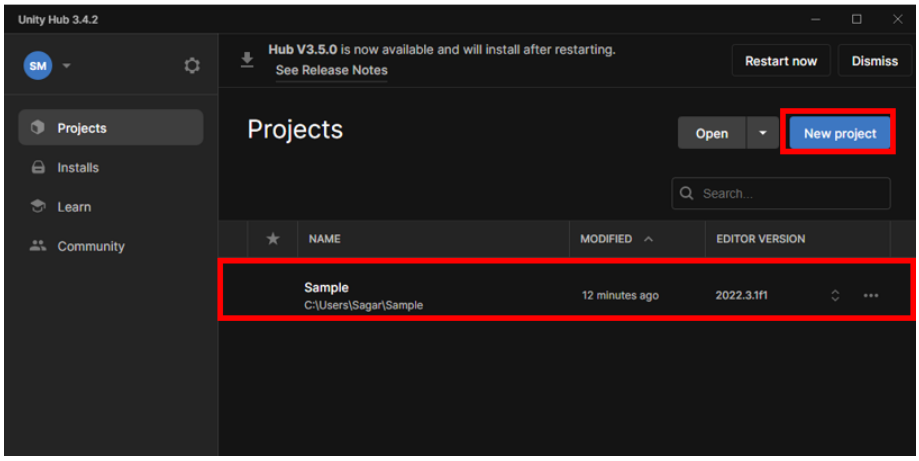
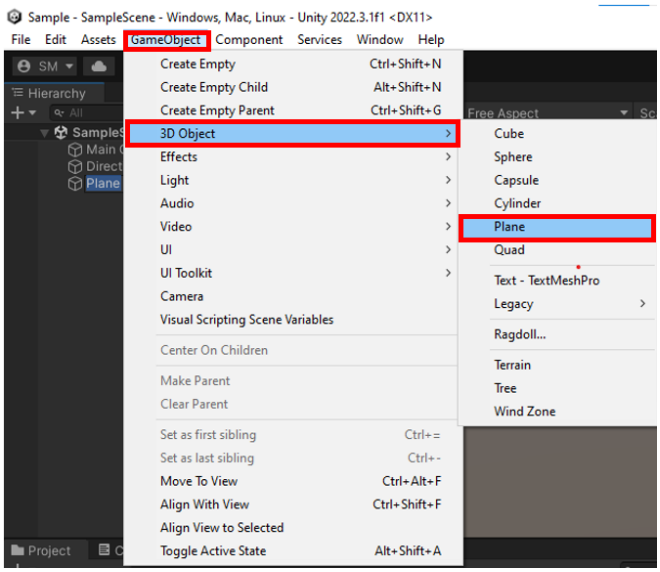
Screenshots:

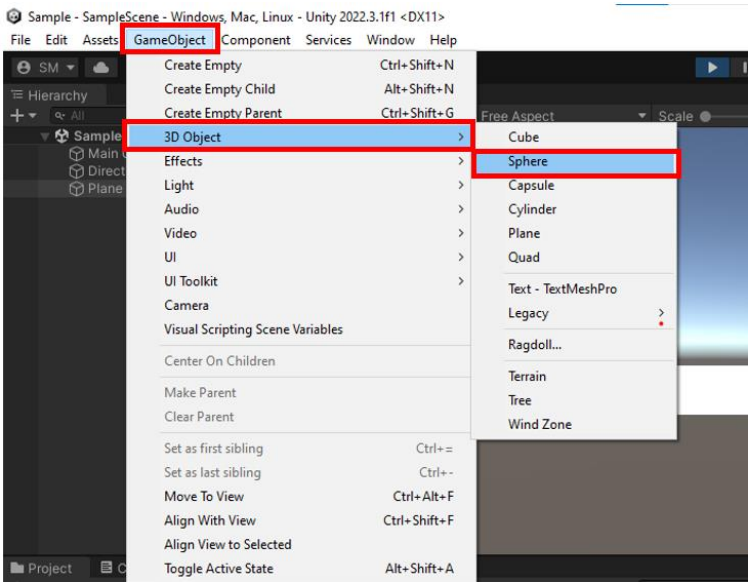
Step 1 to 3



Step 4:



Step 5:



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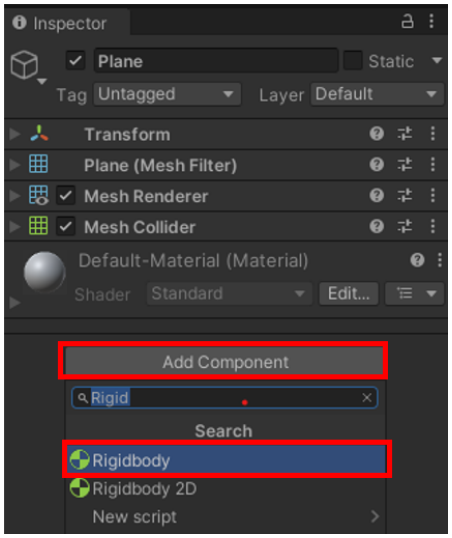
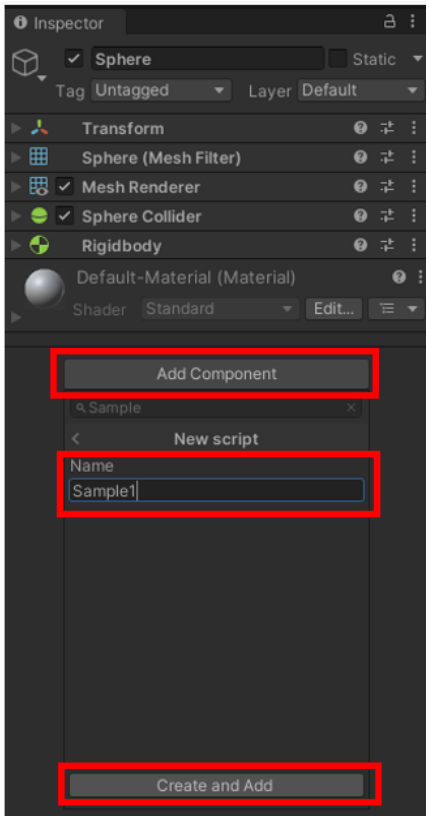
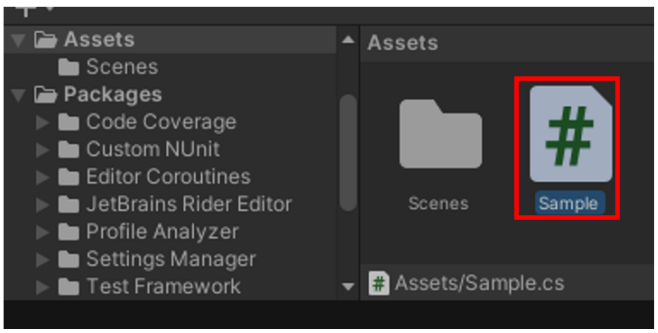


Step 7:



Step 8:



Step 9:

C# Code

```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;

public class light : MonoBehaviour
{
    public Light myLight;
    void Start()
    {
        myLight.GetComponent<Light>();
    }
    // Update is called once per frame
    void Update()
    {
        if (Input.GetKey(KeyCode.L))
        {
            myLight.enabled = !myLight.enabled;
        }
    }
}
```

Step 10:

Run the file in Unity3D and see the output.

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Practical 7

Aim: Implementing virtual environment to Move along with camera

Step No.	Step
1.	Open Unity3D
2.	Click on New Project --> Select 'All Templates' --> Select 3D Core --> Give a Project Name --> Click on Create Project
3.	Double Click on the Project Created
4.	Select Game Object --> 3D Object --> Plane
5.	Select Game Object --> 3D Object --> Sphere
6.	Now Click on 'Add Component' --> Select 'Rigid Body'
7.	Now Click on 'Add Component' --> Type a **new name** --> Select Script --> Select Create and Add [Repeat the process twice – since we need to create 2 scripts]
8.	Double Click on the Script File Created
9.	Type the code in the file that opens:

Movement.cs

```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;

public class movement : MonoBehaviour
{
    public float speed = 0.1f;
    // Start is called before the first frame update
    void Start()
    {

    }

    // Update is called once per frame
    void Update()
    {
        if (Input.GetKey(KeyCode.D))
        {
            transform.Translate(50f * speed * Time.deltaTime, 0, 0);
        }
        if (Input.GetKey(KeyCode.W))
        {
            transform.Translate(0, 0, 50f * speed * Time.deltaTime);
        }
        if (Input.GetKey(KeyCode.A))
        {
            transform.Translate(-50f * speed * Time.deltaTime, 0,
0);
        }
        if (Input.GetKey(KeyCode.S))
        {
            transform.Translate(0, 0, -50f * speed *
Time.deltaTime);
        }

    }
}
```

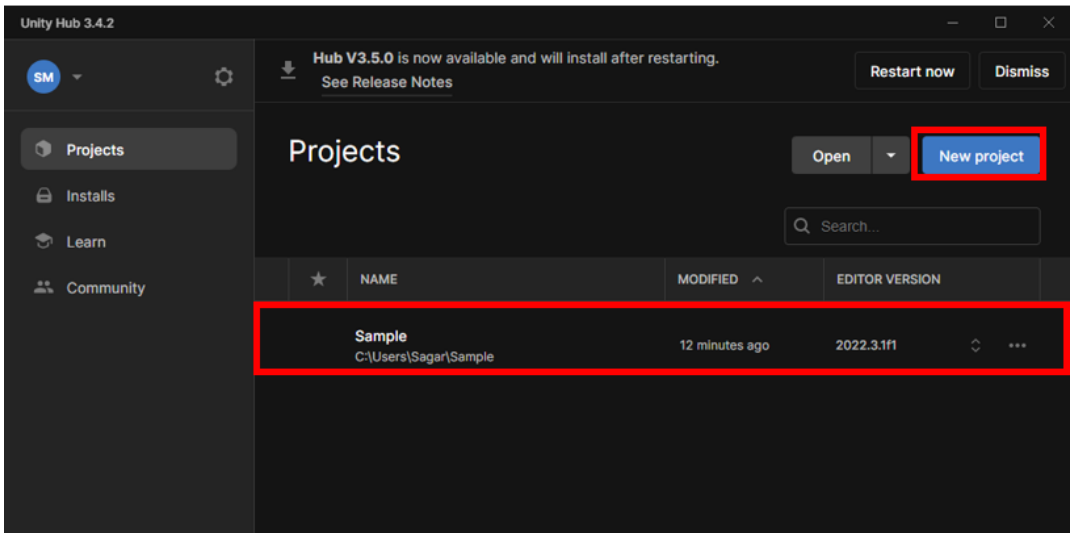
Camerafollow.cs

#Create the camerafollow.cs script in 'Main Camera'.

```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;

public class camerafollow : MonoBehaviour
{
   public GameObject player;
    public Vector3 offset;
    void Start()
    {
        offset = transform.position - player.transform.position;
    }
    // Update is called once per frame
    void Update()
    {

    }
    void LateUpdate()
    {
        transform.position = player.transform.position + offset;
    }

}
```

Screenshots:

Step 1 to 3



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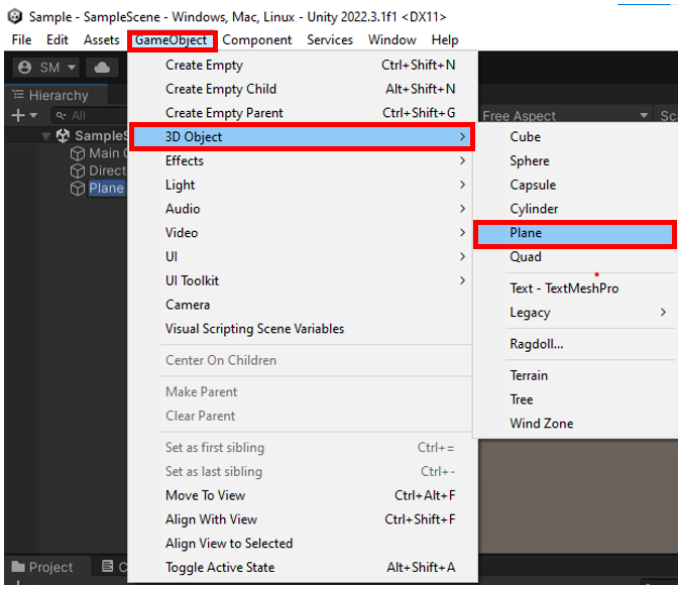
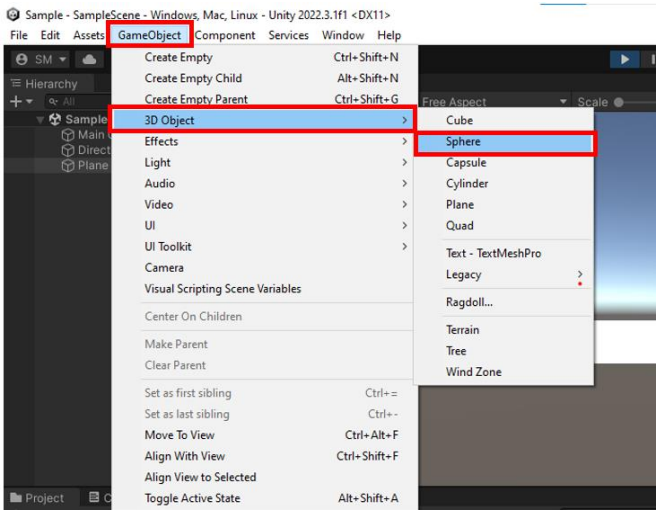
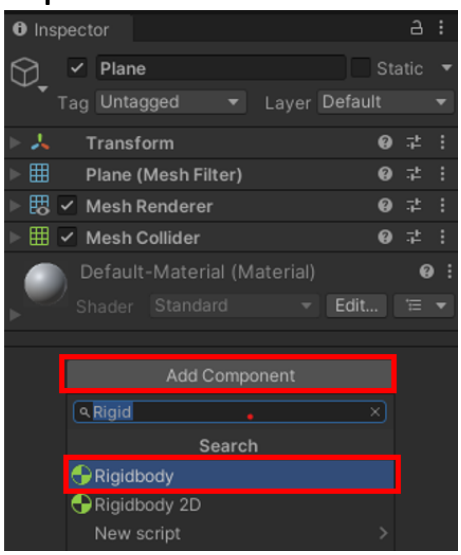
Step 4:



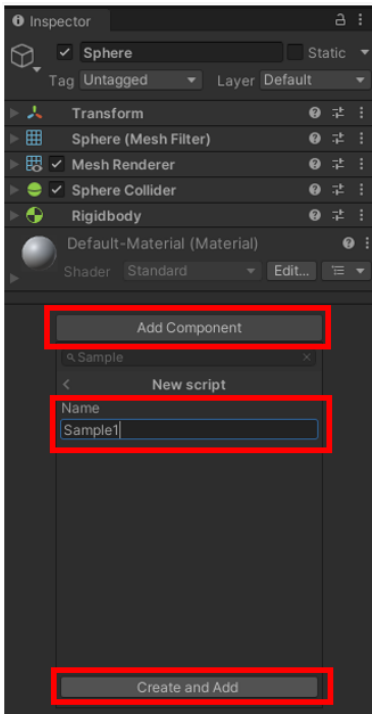
Step 5:



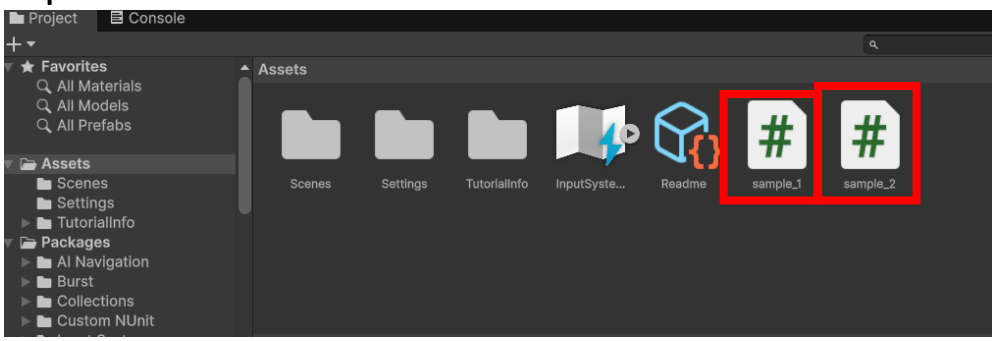
Step 6:

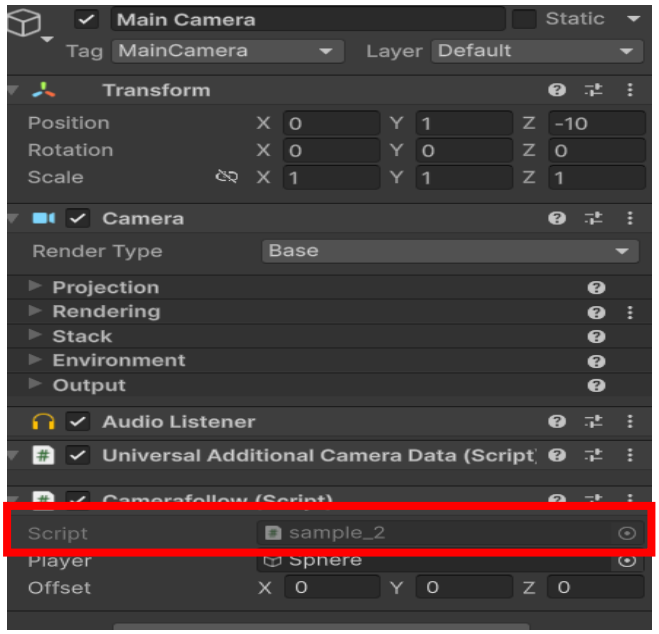


Step 7:


Inspector
Sphere
Static
Tag Untagged
Layer Default
Transform
Sphere (Mesh Filter)
Mesh Renderer
Sphere Collider
Rigidbody
Default-Material (Material)
Shader Standard
Edit...
Add Component
Sample
New script
Name
Sample1
Create and Add

Step 8:


Project
Console
Favorites
All Materials
All Models
All Prefabs
Assets
Scenes
Settings
TutorialInfo
Packages
AI Navigation
Burst
Collections
Custom NUnit
Assets
Scenes
Settings
TutorialInfo
InputSyste...
Readme
sample_1
sample_2


Main Camera
Static
Tag MainCamera
Layer Default
Transform
Position X 0 Y 1 Z -10
Rotation X 0 Y 0 Z 0
Scale X 1 Y 1 Z 1
Camera
Render Type Base
Projection
Rendering
Stack
Environment
Output
Audio Listener
Universal Additional Camera Data (Script)
Script sample_2
Player Sphere
Offset X 0 Y 0 Z 0

Step 9:

C# Code

Movement.cs

```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;

public class movement : MonoBehaviour
{
    public float speed = 0.1f;
    // Start is called before the first frame update
    void Start()
    {
    }
    // Update is called once per frame
    void Update()
    {
        if (Input.GetKey(KeyCode.D))
        {
           transform.Translate(50f * speed * Time.deltaTime, 0, 0);
        }
        if (Input.GetKey(KeyCode.W))
        {
            transform.Translate(0, 0, 50f * speed * Time.deltaTime);
        }
        if (Input.GetKey(KeyCode.A))
        {
            transform.Translate(-50f * speed * Time.deltaTime, 0,
0);
        }
        if (Input.GetKey(KeyCode.S))
        {
            transform.Translate(0, 0, -50f * speed *
Time.deltaTime);
        }
    }
}
```

Camerafollow.cs

#Create the camerafollow.cs script in 'Main Camera'.

```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;

public class camerafollow : MonoBehaviour
{
   public GameObject player;
    public Vector3 offset;
    void Start()
    {
        offset = transform.position - player.transform.position;
    }
    // Update is called once per frame
    void Update()
    {
    }
    void LateUpdate()
    {
        transform.position = player.transform.position + offset;
    }
}
```

Step 10:

Run the file in Unity3D and see the output.

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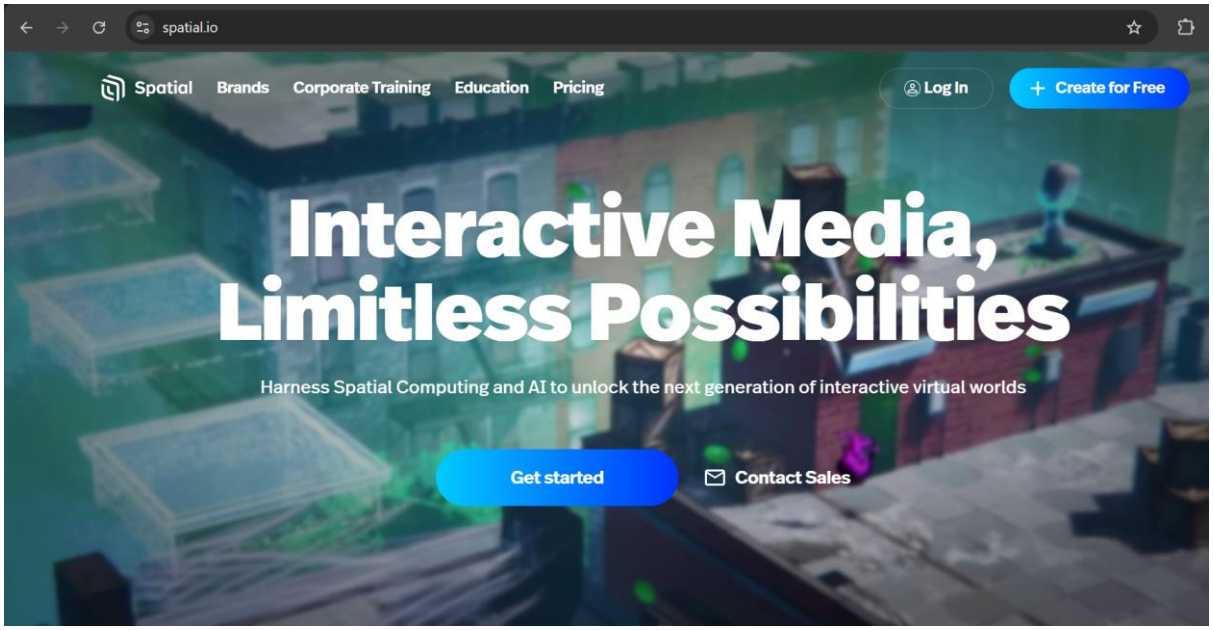
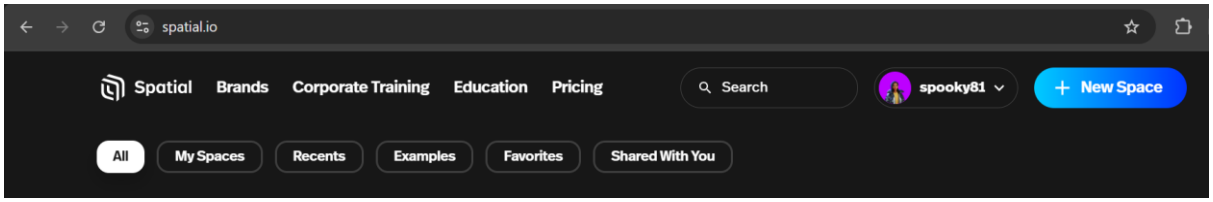
Practical 8: Working with Spatial

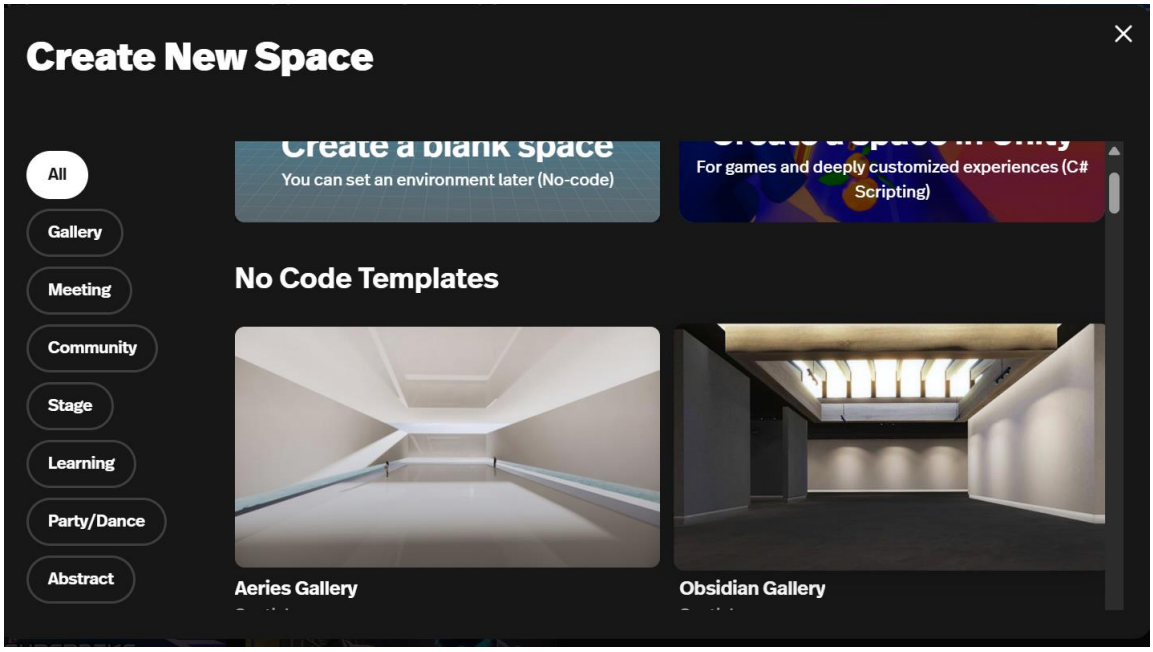
Step 1: Go to https://www.spatial.io/ in Browser and Create Free account

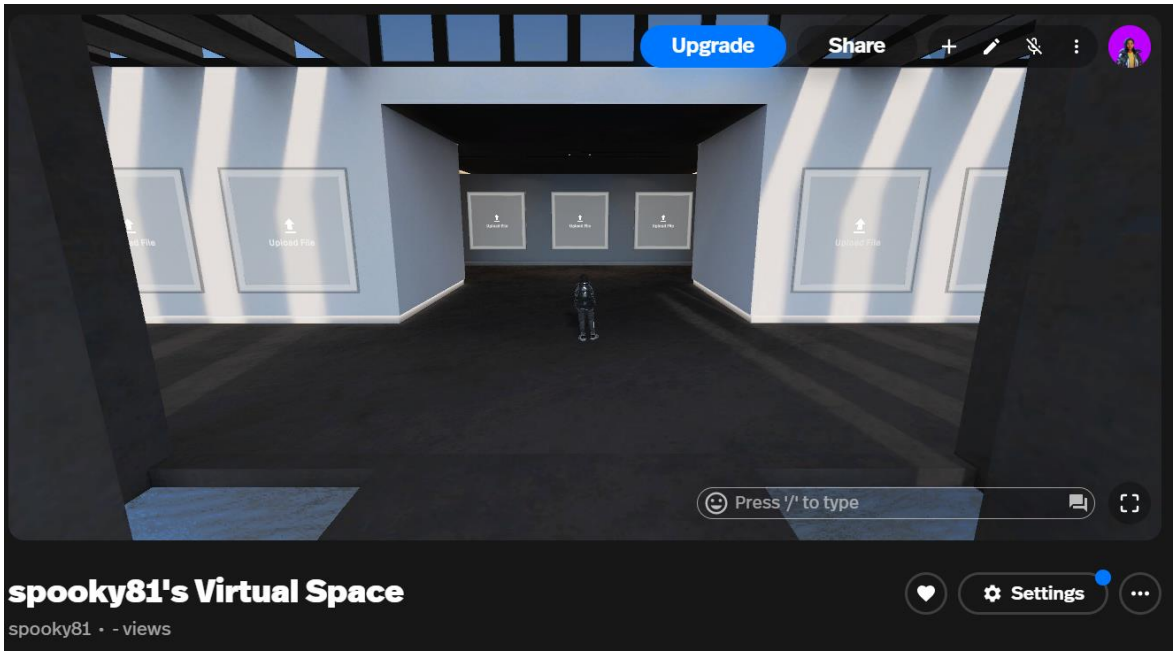


Step 2: Create New Space.

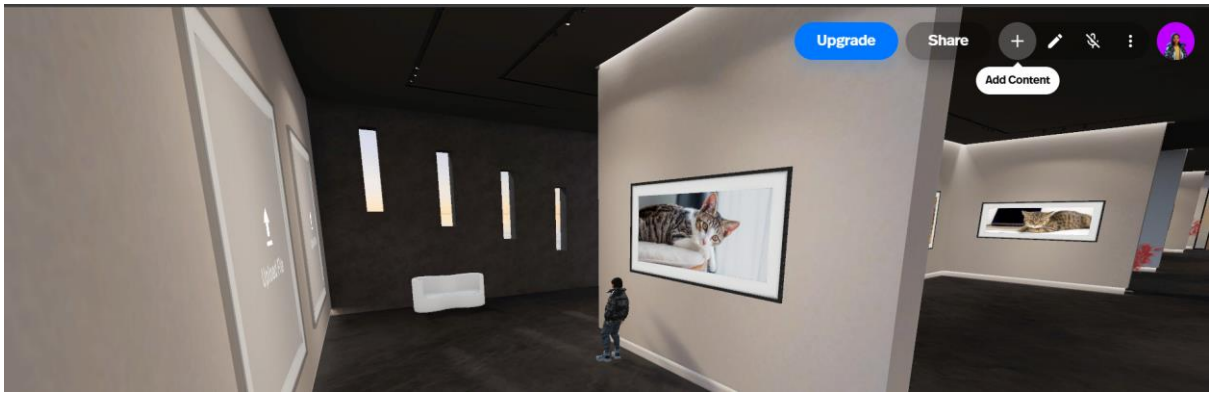


Step 3: Select a Template from 'No Code Templates'.

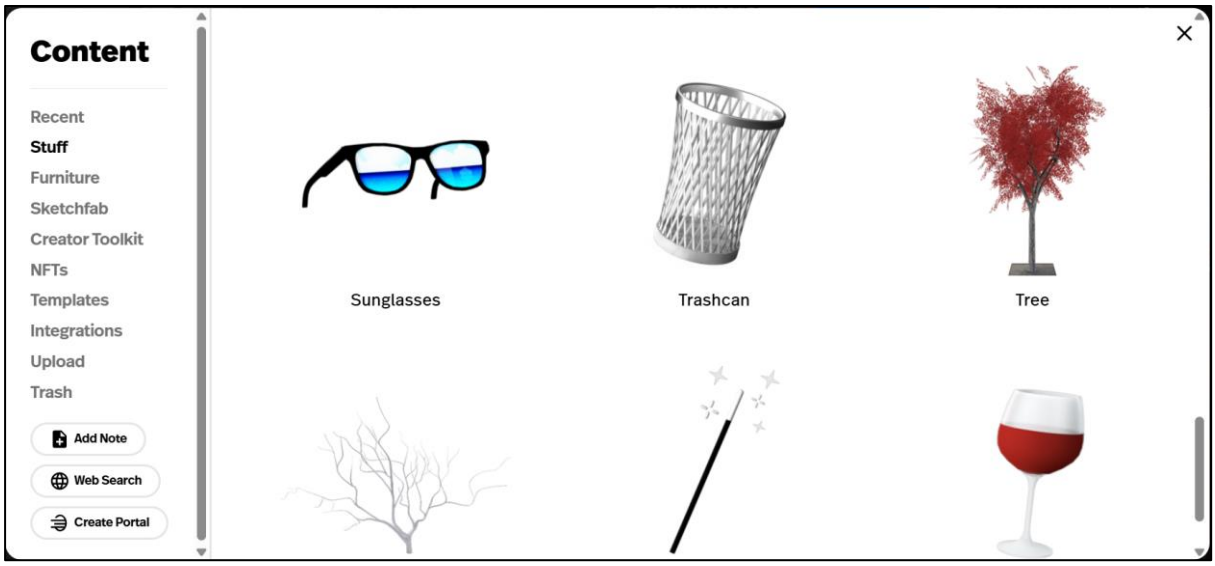




Step 4: Click on Add Content



Step 5: Select Content



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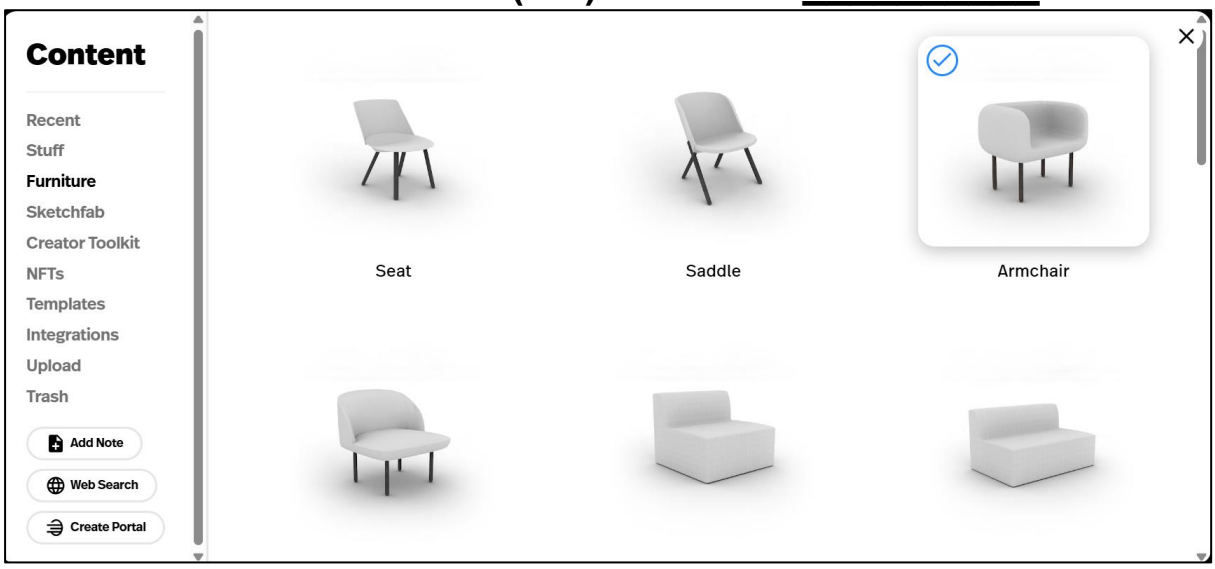
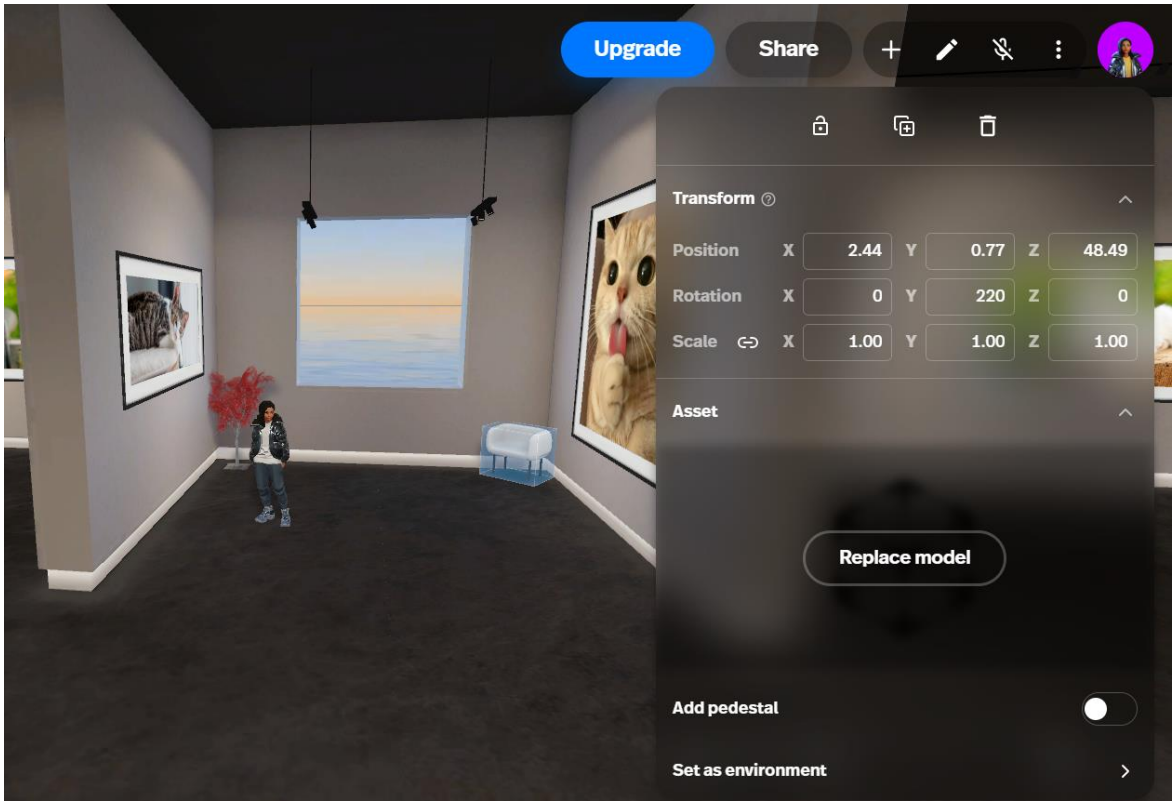
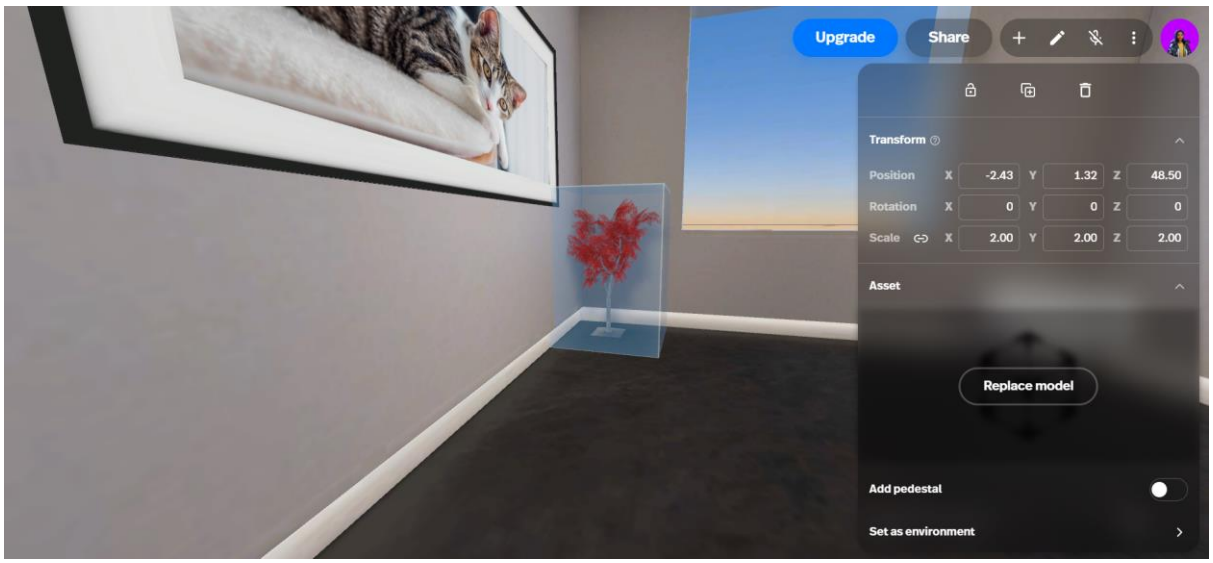


Step 6: Adjust position, rotation and scale accordingly.





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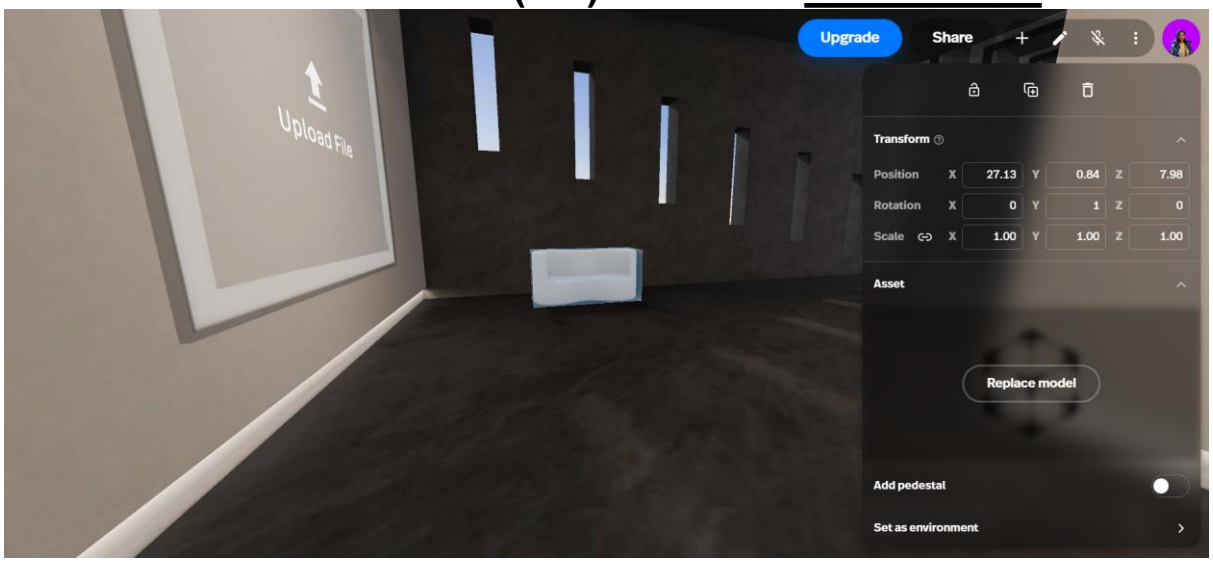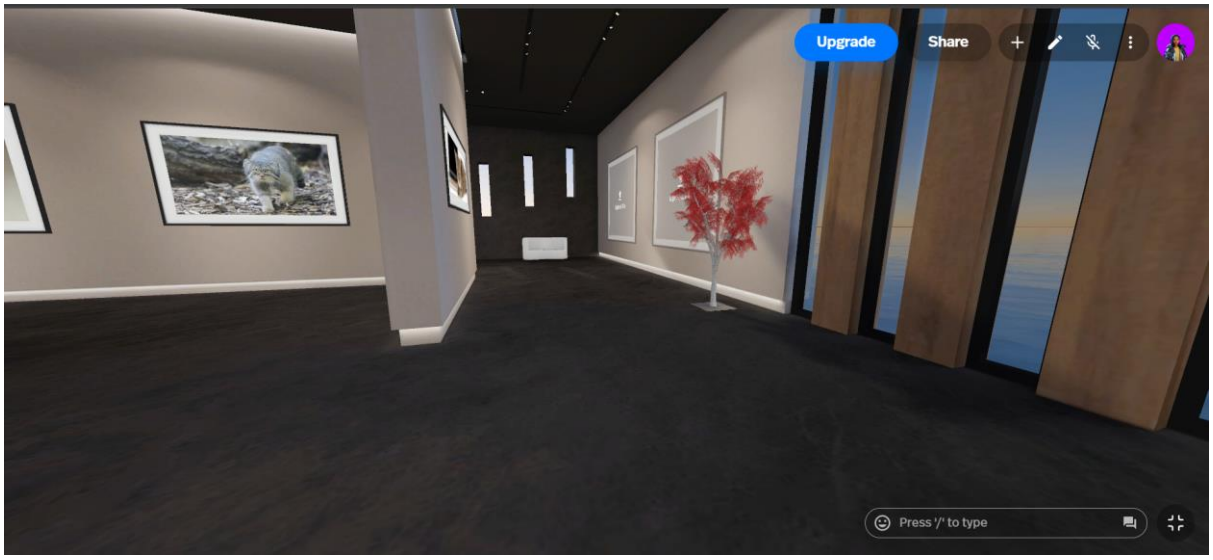
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